Inha University in Tashkent
School of Computer Science and Information Engineering
Course Name: Object Oriented Programming 2
Semester and Year: Spring Freshman, 2026
Laboratory Assignment 3: Object-Oriented Programming in C++

Note: 1. Submission only via E-CLASS is acceptable (If you have any issue with E-CLASS, Please consult IT Dept.)

2. Comments should be included in your source code.

3. Do not plagiarize on doing your work, if we catch student doing so, he/she will get ZERO for the Assignment.

4. Upload your completed laboratory exercise as a zip file including the appropriate files **(.cpp, and .h),** to the eClass system before the deadline. The zip file must be named with your **ID number.**

Exercise 3: Automated Teller Machine (ATM)

Objective: Design a simple ATM simulation system using class composition. The system must manage user account details, perform deposit/withdraw operations, and display user information.

Tasks:

1. Define Three Classes: Account, Balance, ATM. ATM must have a composition relationship with Account and Balance (they are members inside ATM).
2. **CLASS: ACCOUNT**

- Private data members of class Account: bankCard (int), accountName (string), pinNumber (int).
- Static Data Member: static int accountCount. Counts how many Account objects currently exist.

- Define the parameterized constructor initializes bankCard, accountName, pinNumber. Validate: bankCard > 0, pinNumber must be a 4-digit number (1000–9999). If invalid → correct or throw.
- Define const member function: void display() const. It shows account details (bankCard, accountName, pinNumber).
- Define Static Member Function: static int getAccountCount().

3. **CLASS: BALANCE**

- Private member of class Balance: balance (double).
- Static Data Member: static int balanceObjects. Counts how many Balance objects exist.
- Define the parameterized constructor initialize balance. If initial balance < 0 → set to 0.
- Define void deposit(double amount); amount must be > 0.
- Define bool withdraw(double amount); amount must be > 0 and <= balance, returns true if success, else false
- Define const member function: void showBalance() const. It show balance details.
- Define Static Member Function: static int getBalanceObjectCount();

4. **CLASS: ATM**

- Private data members of class ATM(composition): Account account and Balance balance.
- Static Data Member: static int atmCount. Counts how many ATM objects exist.
- Define the parameterized constructor must initialize account and balance using initializer list: Account with (card, name, pin) and Balance with (initialBalance).

- Define const member function: void print() const; Must call: account.display(); balance.showBalance();

- Define Static Member Function: static int getATMCount();

5. In main(), allocate a dynamic array of ATM objects.

Output:

```
Enter number of ATM users: 1

 Enter info for user 1
Card number: 12345
Account name: ALI
PIN (4 digits): 1234
Initial balance: 1000


BEFORE TRANSACTIONS
User 1:
Account Details:
  Card Number : 12345
  Name        : ALI
  PIN         : 1234
Balance:
  Current Balance: 1000.00

Enter user index for deposit (1..1): 1
Enter deposit amount: -500
Invalid deposit amount. Must be > 0.

Enter user index for withdrawal (1..1): 2000
Invalid user index.


AFTER TRANSACTIONS
User 1:
Account Details:
  Card Number : 12345
  Name        : ALI
  PIN         : 1234
Balance:
  Current Balance: 1000.00

 STATIC COUNTERS
Total Account objects: 1
Total Balance objects : 1
Total ATM objects     : 1
```